### 9.2.3 Energy and Momentum in Electromagnetic Waves

According to Eq. 8.5, the energy per unit volume in electromagnetic fields is

$$u = \frac{1}{2}\left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2\right). \tag{9.54}$$

In the case of a monochromatic plane wave (Eq. 9.49 or 9.53),[9]

$$B^2 = \frac{1}{c^2} E^2 = \mu_0 \epsilon_0 E^2, \tag{9.55}$$

so the *electric and magnetic contributions are equal*:

$$u = \epsilon_0 E^2 = \epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta). \tag{9.56}$$

As the wave travels, it carries this energy along with it. The energy flux density (energy per unit area, per unit time) transported by the fields is given by the Poynting vector (Eq. 8.10):

$$\mathbf{S} = \frac{1}{\mu_0}(\mathbf{E} \times \mathbf{B}). \tag{9.57}$$

For monochromatic plane waves propagating in the $z$ direction,

$$\mathbf{S} = c\epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta)\hat{\mathbf{z}} = cu\hat{\mathbf{z}}. \tag{9.58}$$

Notice that $\mathbf{S}$ is the energy density ($u$) times the velocity of the waves ($c\hat{\mathbf{z}}$) – as it *should* be. For in a time $\Delta t$, a length $c\Delta t$ passes through area $A$ (Fig. 9.12), carrying with it an energy $uAc\Delta t$. The energy per unit time, per unit area, transported by the wave is therefore $uc$.

Electromagnetic fields not only carry energy, they also carry momentum. In fact, we found in Eq. 8.29 that the momentum density stored in the fields is

$$\mathbf{g} = \frac{1}{c^2}\mathbf{S}. \tag{9.59}$$

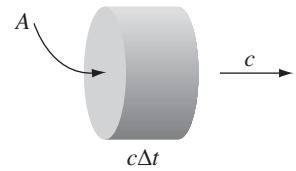



Fig. 9.12

[9] Notice that I'm using the *real* fields to calculate energy density. You might be tempted to introduce

$$\tilde{u} \equiv \tfrac{1}{2}(\epsilon_0 \tilde{E}^2 + \tfrac{1}{\mu_0}\tilde{B}^2),$$

and take its real part, but

$$\mathrm{Re}(\tilde{u}) = \epsilon_0 E_0^2 \cos[2(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta)] \neq u.$$

Why doesn't this work?

For monochromatic plane waves, then,

$$\mathbf{g} = \frac{1}{c}\epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta)\hat{\mathbf{z}} = \frac{1}{c}u\hat{\mathbf{z}}. \tag{9.60}$$

In the case of *light*, the wavelength is so short ($\sim 5 \times 10^{-7}$ m), and the period so brief ($\sim 10^{-15}$ s), that any macroscopic measurement will encompass many cycles. Typically, therefore, we're not interested in the rapidly fluctuating cosine-squared term in the energy and momentum densities; all we want is the *average* value. Now, the average of cosine-squared over a complete cycle[10] is 1/2, so

$$\langle u \rangle = \frac{1}{2}\epsilon_0 E_0^2, \tag{9.61}$$

$$\langle \mathbf{S} \rangle = \frac{1}{2}c\epsilon_0 E_0^2 \hat{\mathbf{z}}, \tag{9.62}$$

$$\langle \mathbf{g} \rangle = \frac{1}{2c}\epsilon_0 E_0^2 \hat{\mathbf{z}}. \tag{9.63}$$

I use brackets, $\langle\ \rangle$, to denote the (time) average over a complete cycle (or *many* cycles, if you prefer). The average power per unit area transported by an electromagnetic wave is called the **intensity**:

$$I \equiv \langle S \rangle = \frac{1}{2}c\epsilon_0 E_0^2. \tag{9.64}$$

When light falls (at normal incidence) on a perfect absorber, it delivers its momentum to the surface. In a time $\Delta t$, the momentum transfer is (Fig. 9.12) $\Delta \mathbf{p} = \langle \mathbf{g} \rangle A c \Delta t$, so the **radiation pressure** (average force per unit area) is

$$P = \frac{1}{A}\frac{\Delta p}{\Delta t} = \frac{1}{2}\epsilon_0 E_0^2 = \frac{I}{c}. \tag{9.65}$$

(On a perfect *reflector* the pressure is *twice* as great, because the momentum switches direction, instead of simply being absorbed.) We can account for this pressure qualitatively, as follows: The electric field (Fig. 9.10) drives charges in the $x$ direction, and the magnetic field then exerts on them a force $q(\mathbf{v} \times \mathbf{B})$ in the $z$ direction. The net force on all the charges in the surface produces the pressure.[11]

**Problem 9.10.** The intensity of sunlight hitting the Earth is about 1300 W/m$^2$. If sunlight strikes a perfect absorber, what pressure does it exert? How about a perfect reflector? What fraction of atmospheric pressure does this amount to?

[10] There is a cute trick for doing this in your head: $\sin^2\theta + \cos^2\theta = 1$, and over a complete cycle the average of $\sin^2\theta$ is equal to the average of $\cos^2\theta$, so $\langle \sin^2 \rangle = \langle \cos^2 \rangle = 1/2$. Formally,

$$\frac{1}{T}\int_0^T \cos^2(kz - 2\pi t/T + \delta)\, dt = \frac{1}{2}.$$

[11] Actually, it's a little more subtle than this – see Prob. 9.11.